PROJECT SYNOPSIS

On

"Online Food Ordering System"

Towards partial fulfillment of the requirement for the award of degree of

Master of Computer Application

From

Integral University

Lucknow

Academic Session 2023 – 2024

Department of Computer Application

Submitted By:

Ayan Ahmad (2200101284)

Sohaim Aslam (1900103491)

Under Guidance of:

Mr. Khalid Jamal

(Assistant Professor)

Dept. of Computer Application

1. Title of the Project: “Online Food Ordering System”

Description of Project:

The advent of digital technology has transformed various sectors, including the food industry. E-Foodie is a comprehensive online food ordering system designed to streamline the process of ordering food from restaurants, ensuring convenience and efficiency for both customers and restaurant owners. This project aims to bridge the gap between food enthusiasts and their favorite eateries by providing a user-friendly platform for placing orders online.

The main purpose of an online food ordering system is to revolutionize the way people interact with food establishments, offering a seamless and convenient platform for customers to order meals from a variety of restaurants without the constraints of physical location or time. By leveraging digital technology, these systems aim to simplify the entire ordering process, from menu browsing to payment, providing customers with a hassle-free experience. Moreover, online food ordering systems empower restaurants to efficiently manage their operations, streamline order processing, and expand their reach to a wider customer base. Ultimately, the primary goal is to enhance accessibility, convenience, and satisfaction for both customers and restaurant owners in the increasingly digitalized food service industry.

2. Objective:

1. Nominal charge of 15% for declining food delivery: Customers may incur a fee of 15% of their order total if they choose to decline a food delivery.

2. Purpose of the charge: This fee may serve multiple purposes, including covering administrative costs associated with processing orders and coordinating deliveries. It may also discourage frequent cancellations, which can disrupt restaurant operations and inconvenience delivery personnel.

3. Industry standard: Such charges are common in the online food ordering industry and are often included in the terms of service of major platforms.

Charging for unoccupied cash on delivery orders helps cover costs incurred by the restaurant and delivery personnel. Additionally, being blacklisted or restricted from further orders may be a consequence for users who frequently cancel orders, as it helps maintain order integrity and ensures fair service for all users.

*Our project is different from Zomato and Swiggy such that we will showcase on nominal charge of 15% for declining food orders.

3. Features:

- User Registration and Login:
  Allow customers to create accounts and log in securely.
  Store user preferences and order history for personalized experiences.
- Menu Browsing:
  Provide an intuitive interface for browsing restaurant menus.
  Allow filtering by cuisine, price, dietary preferences, etc.
- Order Placement:
  Enable customers to add items to their cart and customize orders.
  Offer seamless checkout processes with multiple payment options.
- Payment Gateway Integration:
  Securely process payments through various methods like credit/debit cards, digital wallets, and cash on delivery.
- Order Tracking:
  Allow customers to track the status of their orders in real-time.
  Provide updates on order confirmation, preparation, and delivery.
- Feedback and Ratings:
  Allow customers to leave reviews and ratings for restaurants and dishes.
  Enable restaurant owners to gather feedback and improve service quality.
- Restaurant Management:
  Provide restaurant owners with dashboards to manage menus, track orders, and monitor sales.
  Enable easy updates to menus and promotional offers.
- Promotions and Discounts:
  Integrate promotional offers, discounts, and loyalty programs to incentivize purchases.
  Notify customers of special deals and limited-time offers.
- Geolocation Services:
  Utilize mapping and geolocation services to suggest nearby restaurants.
  Provide accurate delivery estimates based on customer location.
- Customer Support:
  Offer channels for customer support such as live chat, email, or phone.
  Resolve queries and issues promptly to ensure customer satisfaction.
- Covering Administrative Costs:
  The charge may help cover administrative expenses incurred by the platform when processing orders, managing logistics, and coordinating with restaurants and delivery personnel.
- Discouraging Frequent Cancellations:
  By imposing a charge for declining food delivery, the platform aims to discourage users from canceling orders frequently.
- Encouraging Responsible Ordering Behavior:
  Implementing a charge for declining food delivery encourages users to be more thoughtful and considerate when placing orders.

- Ensuring Fair Service for All Users:
  By discouraging frequent cancellations, the platform can ensure fair and equitable service for all users.

4. Methodology:

4.1. Admin Workflow Process:

User goes to home page of domain. If the user has an account then they can login to the food management system otherwise they need to register for an account to login in homepage.

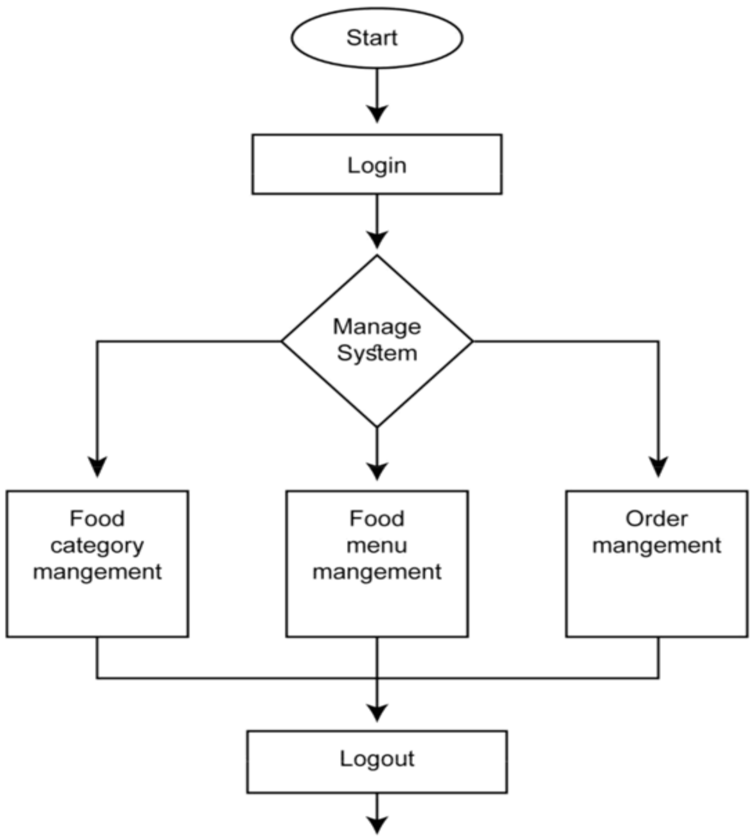

Figure 4.1: Admin Workflow Process

4.2. Customer Workflow Process:

Initially to visit the food categories or menu, users do not need to login/register an account. After checking out the categories and menu items, if the user finds his or her desired menu and if they want to order that particular item they can go to order page.

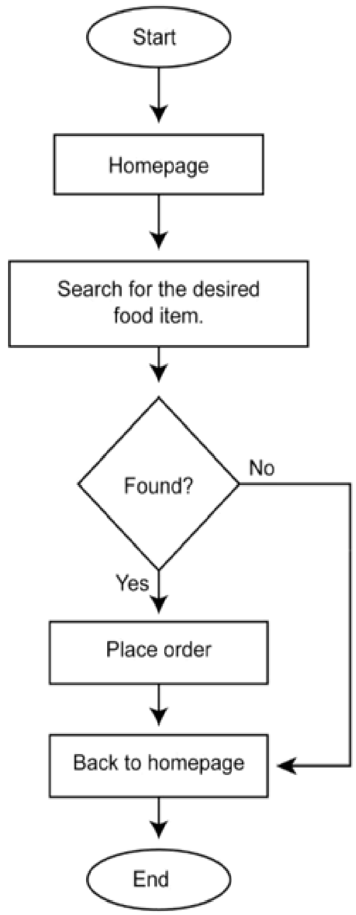

Figure: 4.2: Customer Workflow Process

4.3. Diagrams:

4.3.1. E-R Diagram:

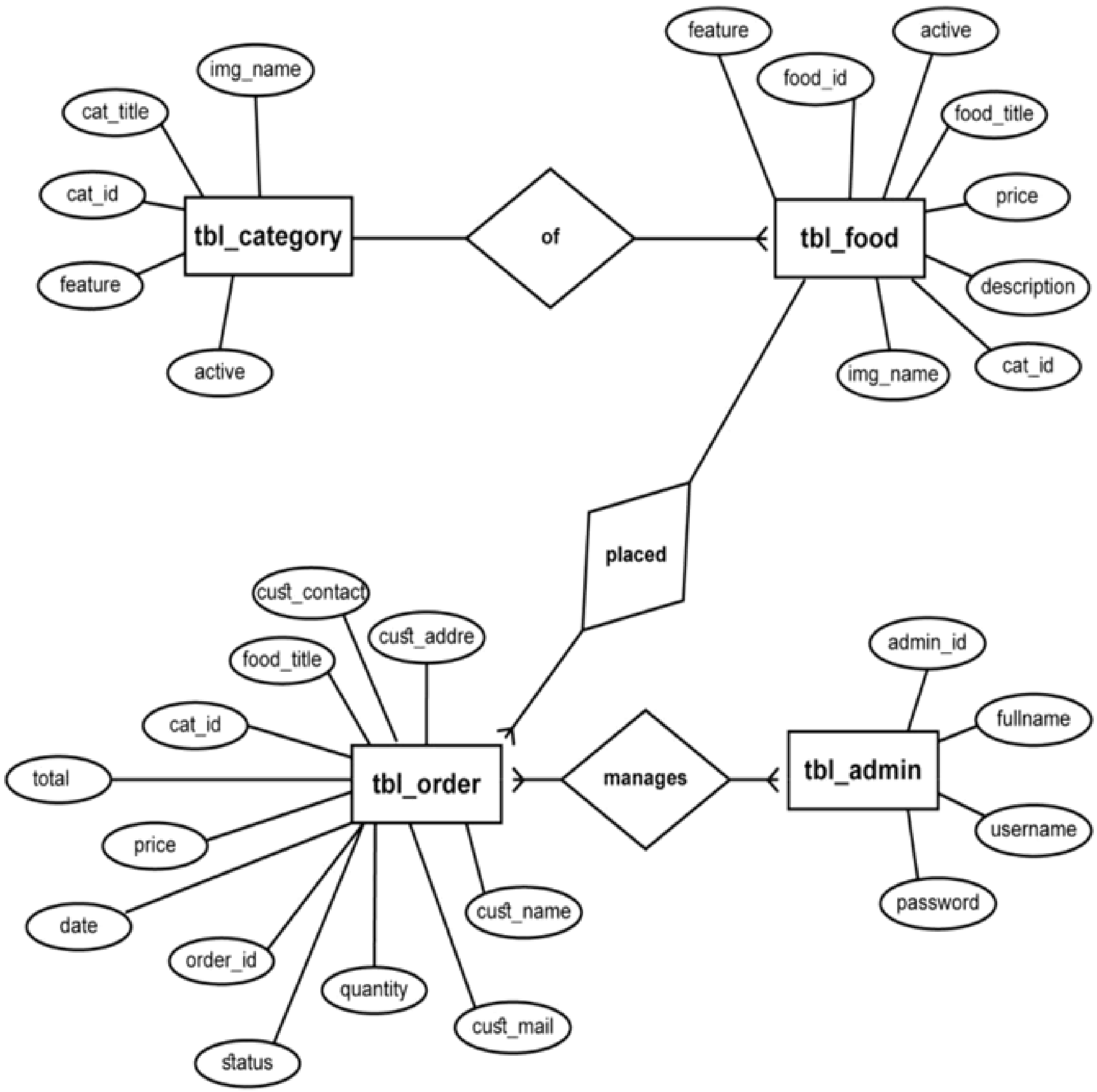

Figure 4.3.1: E-R Diagram

4.3.2. Schema Diagram:

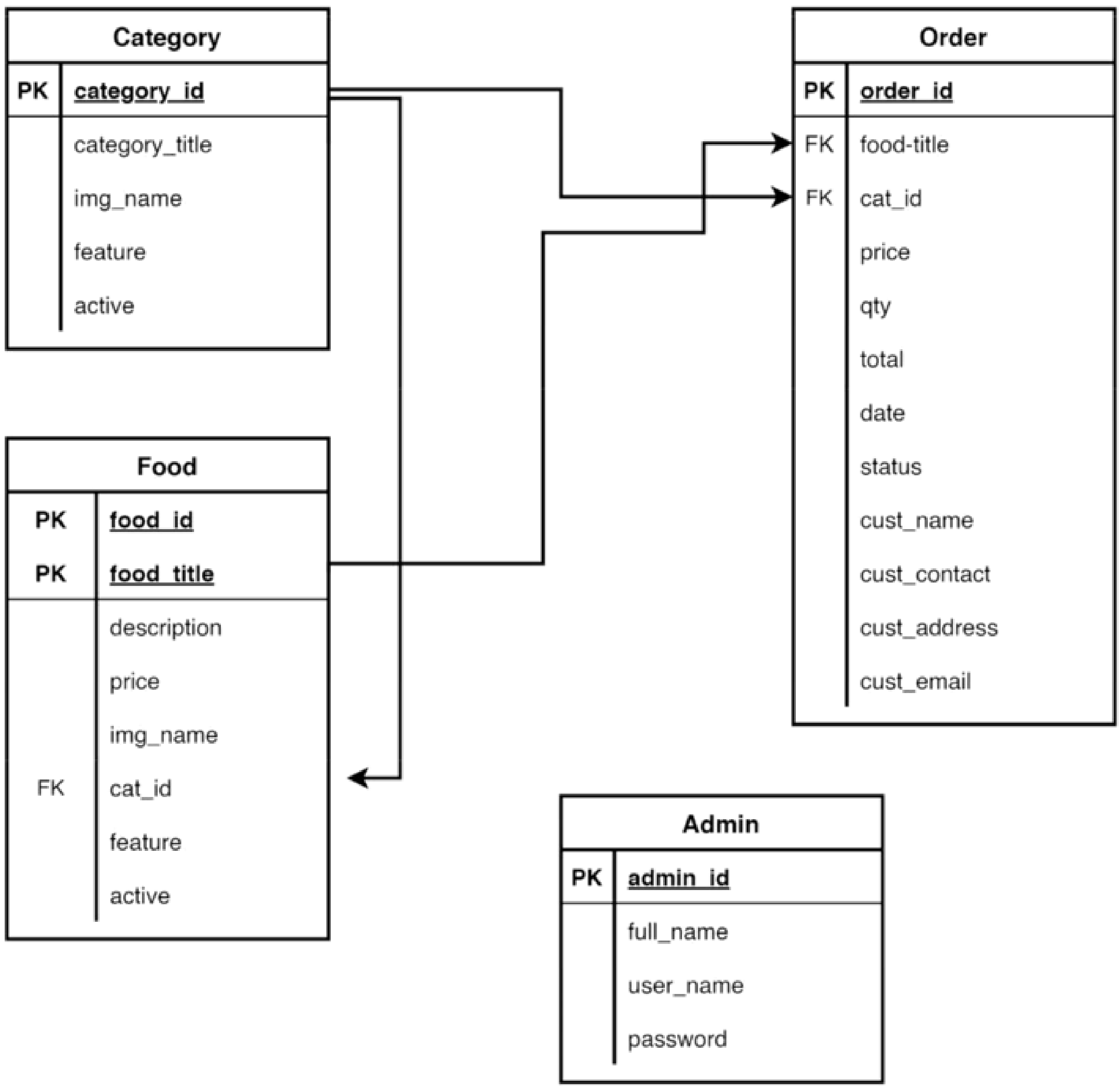

Figure 4.3.2: Schema Diagram

4.3.3. DFD Diagram

Level-0 (Context Level):

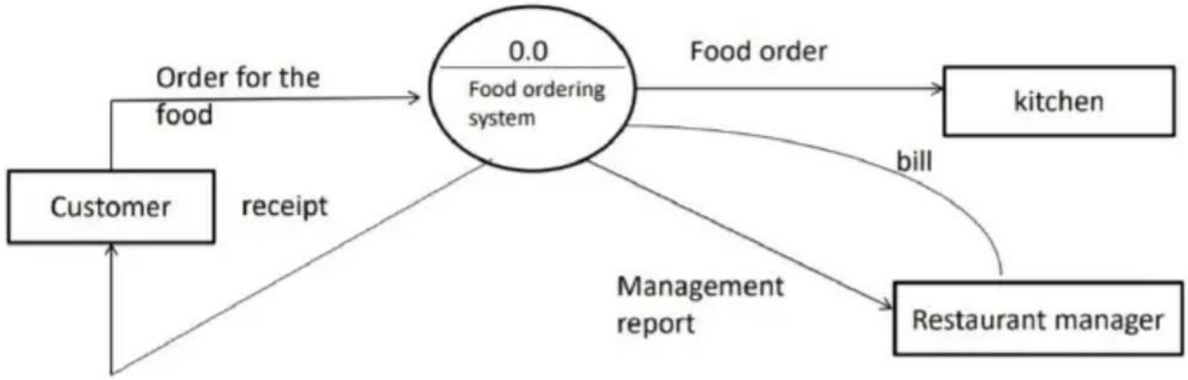

Figure 4.3.3 (i): DFD Diagram (Level-0)

Level-1:

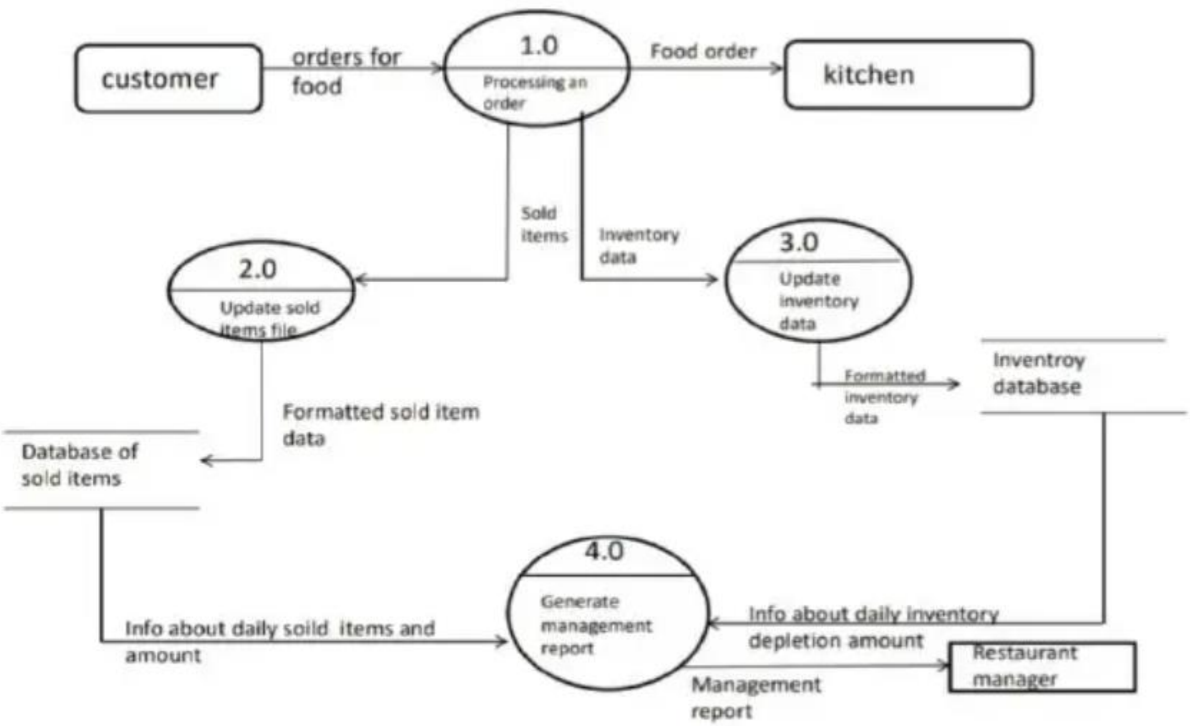

Figure 4.3.3 (ii): DFD Diagram (Level-1)

Level-2:

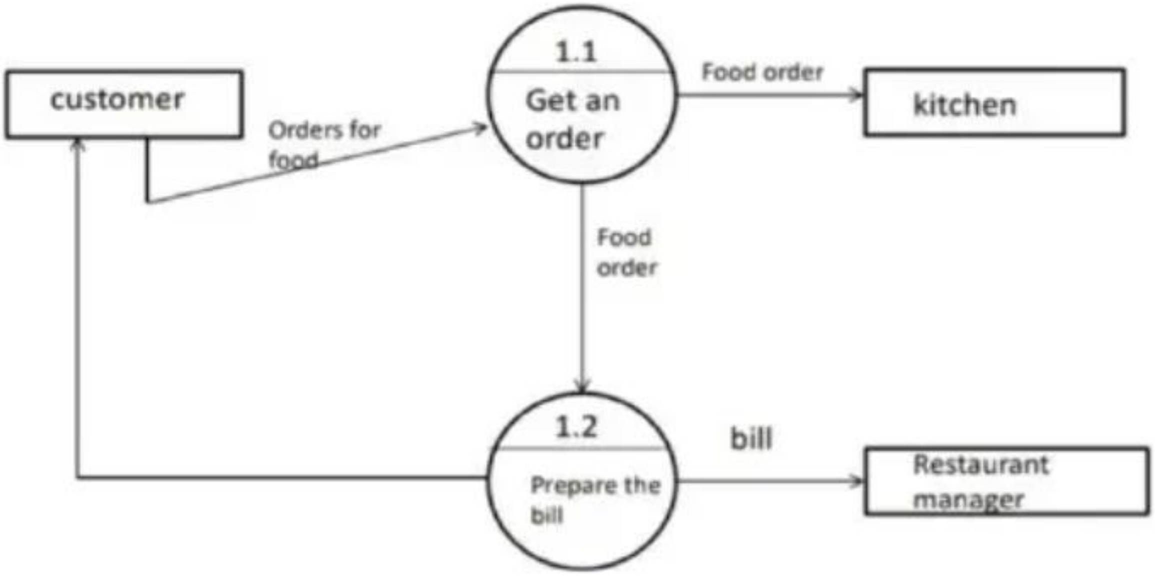

Figure 4.3.3 (iii): DFD Diagram (Level-2)

Level-3:

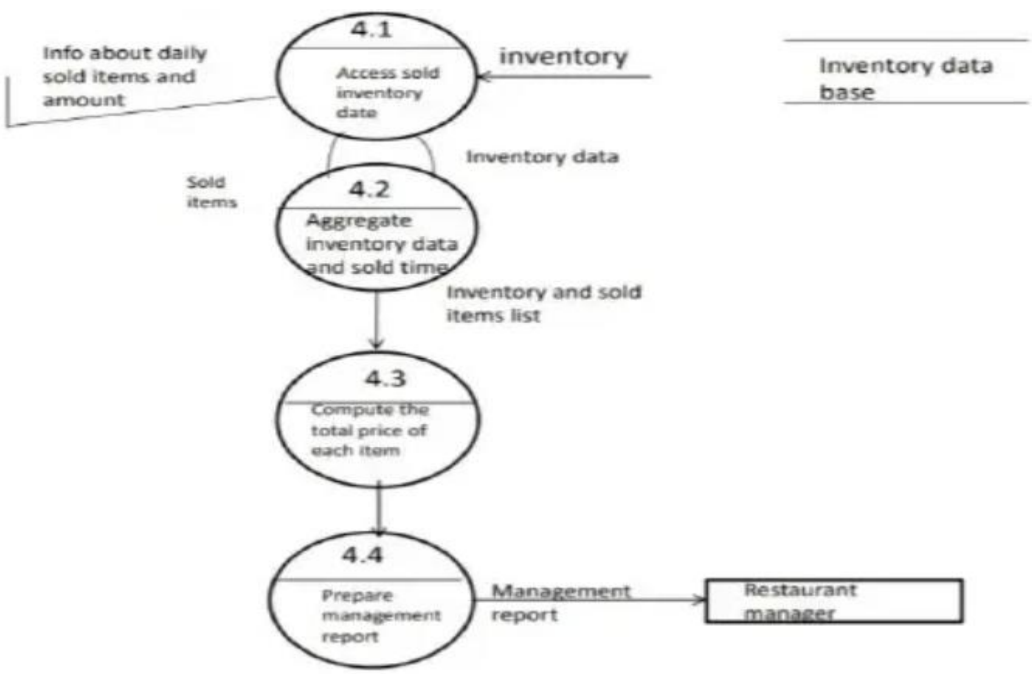

Figure 4.3.3 (iv): DFD Diagram (Level-3)

5. Resources (Hardware and Software) to be used:

Hardware Resources:

Server: The server should have sufficient processing power, memory, and storage capacity to handle user requests, database operations, and other computational tasks.

Storage: Adequate storage space is necessary to store user data, trip information, destination details, and other relevant content. This can be in the form of hard disk drives (HDDs) or solid-state drives (SSDs), depending on performance and scalability requirements.

Backup Systems: Regular backups of data should be performed to prevent data loss in case of hardware failures or other unforeseen events. Backup systems such as external hard drives, cloud storage services, or backup servers are necessary to store redundant copies of critical data.

Software Resources:

Operating System: A server-grade operating system such as Linux (e.g., Ubuntu, CentOS) or Windows Server is required to run the web server software and other necessary applications.

Web Server: Software like Apache HTTP Server or Nginx will be used to serve web pages, handle HTTP requests, and communicate with clients over the internet.

Database Management System (DBMS): A relational database management system (RDBMS) such as MySQL will be used to store and manage structured data related to users, meals, bookings, deliveries, etc.

6. Tools and Techniques:

i. Php

PHP, or Hypertext Preprocessor, is a powerful server-side scripting language primarily used for web development. It's widely popular due to its simplicity, flexibility, and compatibility with various databases and web servers. PHP scripts are embedded within HTML, allowing developers to create dynamic and interactive web pages.

ii. MySQL

MySQL is an open-source relational database management system (RDBMS) that is widely used for storing and managing structured data. It is renowned for its speed, reliability, and ease of use, making it a popular choice for both small-scale and large-scale applications. MySQL employs a client-server architecture, where clients interact with the database server using the SQL (Structured Query Language) to perform various operations such as querying, updating, and managing data.

iii. HTML

HTML, or Hypertext Markup Language, is the standard markup language used for creating and structuring web pages. It provides a set of predefined elements, tags, and attributes that define the structure and content of a web document. HTML documents consist of a series of elements, each enclosed within opening and closing tags, which define the structure and semantics of the content. These elements can include headings, paragraphs, lists, links, images, forms, and more. HTML is used for industry-standard markup language for developing web apps and pages.

iv. CSS

CSS, or Cascading Style Sheets, is a style sheet language used for describing the presentation and appearance of HTML documents. It enables web developers to control the layout, formatting, and visual styling of web pages across different devices and screen sizes. CSS works by applying styling rules to HTML elements, specifying properties such as colors, fonts, margins, padding, and positioning. These styling rules can be defined inline within HTML elements, embedded within HTML documents, or stored externally in separate CSS files for easier maintenance and organization.

v. Java Script

JavaScript is a versatile and powerful programming language primarily used for adding interactivity and dynamic behavior to web pages. It is an essential component of modern web development, enabling developers to create interactive elements, manipulate the content and structure of web pages, and respond to user actions in real-time. JavaScript runs client-side, meaning it executes directly within the web browser, allowing for seamless interaction without the need to reload the entire page.

vi. Visual Studio Code

Visual Studio Code (VS Code) is a free and open-source source code editor developed by Microsoft. It is widely popular among developers for its lightweight yet powerful features, flexibility, and extensive ecosystem of extensions. VS Code provides a modern and customizable user interface with features like syntax highlighting, code completion, debugging support, version control integration, and built-in terminal. It supports a wide range of programming languages and frameworks out of the box, and its rich extension marketplace allows developers to further enhance their workflow with additional functionalities and tools.

vii. XAMPP

XAMPP is a free and open-source cross-platform web server solution stack package developed by Apache Friends. The acronym stands for "X" (meaning cross-platform), "Apache" (the web server), "MySQL" (the database), "PHP" (the programming language), and "Perl" (another programming language). XAMPP allows users to easily set up a local web server environment on their computer, facilitating web development and testing. It includes all the necessary components to run dynamic web applications, such as Apache HTTP Server, MySQL database server, PHP, and Perl interpreters.

viii. GitHub

GitHub is a web-based platform and hosting service that provides version control and collaboration features for software development projects. It allows developers to store, manage, and track changes to their codebase using Git, a distributed version control system. GitHub offers a range of features including code hosting, issue tracking, pull requests, code review, project management, and continuous integration. It enables developers to work collaboratively on projects, contributing code, reviewing changes, and resolving issues together.

7. Project Schedule Plan:

Creating a project schedule plan for the Online Food Ordering System involves breaking down the project into smaller tasks, estimating the time required for each task, and organizing these tasks into a timeline. Here's a simplified example of a project schedule plan:

- Phase 1: Planning and Requirements Gathering (2 weeks)
  Define project scope and objectives (2 days)
  Gather requirements from stakeholders (3 days)
  Create project schedule and task breakdown (2 days)
  Identify resources needed (1 day)
  Finalize project plan and get approval (2 days)

- Phase 2: Design and Architecture (3 weeks)
  Design database schema (3 days)
  Create wireframes/mockups for user interface (5 days)
  Define system architecture and components (3 days)
  Select and set up development tools and environment (2 days)
  Plan integration with external APIs (2 days)
  Review and finalize design and architecture (2 days)

- Phase 3: Development (6 weeks)
  Implement user authentication and registration (2 weeks)
  Develop meal search functionality (2 weeks)
  Implement itineraries and customer review creation feature (2 weeks)
  Integrate booking and delivery management system (3 weeks)
  Develop user profile management module (1 week)
  Implement interactive social sharing (2 weeks)
  Ensure security measures are implemented (2 weeks)

- Phase 4: Testing and Quality Assurance (3 weeks)
  Conduct unit testing for individual components (2 weeks)
  Perform integration testing for the entire system (1 week)
  Identify and fix bugs and issues (1 week)
  Perform security testing and vulnerability assessment (1 week)
  Ensure compliance with usability and accessibility standards (1 week)

8. Project Team:

Sohaim Aslam:

Role: Backend Developer, Database Creation

Responsibilities:

Overseeing the project's progress, ensuring adherence to timelines and requirements.

Leading the development team, writing code, and resolving technical issues.

Managing the database design, optimization, and maintenance.

Conducting quality assurance activities and ensuring the application meets functional and performance requirements.

Ayan Ahmad:

Role: Developer, Frontend Development

Responsibilities:

Collaborating with team member in development tasks.

Focusing on frontend development, including user interface design and implementation.

Conducting testing activities, identifying bugs, and assisting in debugging and problem-solving.

Documenting project progress, codebase, and technical specifications for clarity and maintainability.

As a team, Sohaim Aslam and Ayan Ahmad will work together to develop, test, and deploy the Online Food Delivery application, leveraging their respective skills and expertise to ensure its successful completion.

9. Process Description:

In the process description for the Online Food Delivery System project, the development team utilizes the Visual Studio IDE for HTML development and MySQL for database management. The project leverages Java Servlets for server-side logic and interaction with the frontend, ensuring dynamic and responsive web applications. Additionally, the team employs a Java framework to streamline development tasks and maintain code consistency. By integrating these technologies, the project benefits from a robust development environment, efficient database management, and structured coding practices. This approach enables the team to create a scalable and maintainable online trip planner that meets the needs of users effectively.

10. Conclusion:

In conclusion, the online food ordering system led by students Sohaim Aslam and Ayan Ahmad represents a remarkable achievement in harnessing technology to address real-world challenges. Their innovative solution not only streamlines the food ordering process but also embodies the spirit of entrepreneurship and creativity. By developing a user-friendly platform with features tailored to meet the needs of customers and restaurant owners alike, Sohaim and Ayan have demonstrated their ability to deliver practical solutions that make a meaningful impact. Their dedication, passion, and ingenuity serve as an inspiration to their peers and the broader community, highlighting the potential of young innovators to drive positive change. As their project continues to evolve and gain traction, it is poised to make a significant contribution to the food service industry, enhancing convenience, efficiency, and satisfaction for all stakeholders involved.